Exponential graphs and transformations

1 Consider the functions $y = 2^x$, $y = 3^x$ and $y = 5^x$.

a Sketch the three functions on the same set of axes.

b Determine whether each of the following statements are true:

i None of the curves cross the x-axis.

ii They all have the same y-intercept.

iii All of the curves pass through the point $(1, 2)$.

iv All of the curves have a maximum value.

c State the y-intercept of each curve.

2 Consider the function $y = -(2^x)$.

a Complete the table of values:

x	-5	-4	-3	-2	-1	0	1	2	3	4	5
y											

b Can the value of y ever be zero or positive? Explain your answer.

c Is $y = -(2^x)$ an increasing or decreasing function?

d Describe the behaviour of the function as x increases.

e State the domain.

f State the range.

3 Find the missing coordinate in each ordered pair so that the pair is a solution of $y = -3^x$:

a $(5, \square)$ **b** $\left(\square, -\frac{1}{27}\right)$ **c** $(-1, \square)$ **d** $(\square, -81)$

4 Determine the y-intercept of all exponential functions of the form:

a $y = a^x$ **b** $y = a^{-x}$ **c** $y = -a^x$ **d** $y = -a^{-x}$

5 Consider the function $y = -2.5 \times 4^x$.

a Is $y = -2.5 \times 4^x$ an increasing or decreasing function?

b As x approaches $-\infty$, what value does y approach?

c As x approaches ∞, y what value does y approach?

d State the y-intercept of the graph.

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6 Determine whether following are increasing or decreasing exponential functions:

a $y = 10 \times \left(\frac{3}{5}\right)^x$ **b** $y = 9 \times 3^x$ **c** $y = 3 \times (0.5)^x$ **d** $y = 0.2 \times 2^x$

e $y = 2.2 \times 1.05^x$ **f** $y = 50 \times \left(\frac{7}{4}\right)^x$ **g** $y = 500 \times 0.75^x$ **h** $y = 80 \times 2^{-x}$

i $y = -5 \times 2^x$ **j** $y = -0.5 \times 0.8^x$ **k** $y = -2 \times \left(\frac{1}{4}\right)^x$ **l** $y = -4 \times 3^{-x}$

7 Which of the following graphs of exponential functions rises most steeply?

A $y = 2 \times (2.1)^x$ **B** $y = 2 \times (2.2)^x$ **C** $y = 2 \times (1.7)^x$ **D** $y = 2 \times (1.2)^x$

8 Of the two functions $y = 2^x$ and $y = 3 \times 2^x$, which is increasing more rapidly for $x > 0$?

9 Consider the functions $y = 2^x$ and $y = 2^x - 2$.

a Find the y-intercept of $y = 2^x$.

b Hence, determine the y-intercept of $y = 2^x - 2$.

c State the horizontal asymptote of $y = 2^x$.

d Hence, determine the horizontal asymptote of $y = 2^x - 2$.

10 For each of the following exponential functions:

i Find the y-value of the y-intercept. **ii** Determine the horizontal asymptote.

iii Sketch the graph of the function.

a $y = 3^{-x} - 1$ **b** $y = -3^x - 2$

11 For each of the following exponential functions:

i Find the y-value of the y-intercept. **ii** State the domain.

iii State the range. **iv** Sketch the graph of the function.

a $y = 3^x - 5$ **b** $y = 4^x + 3$

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12 For each of the following exponential functions:

i Find the y-value of the y-intercept.

ii Create a table of values for $-3 \leq x \leq 3$.

iii Determine the horizontal asymptote.

iv Sketch the graph of the function.

iv Is the function increasing or decreasing?

a $y = -3^{-x}$ **b** $y = 3^{2x}$ **c** $y = 3 \times 2^x + 2$

13 Consider the function $y = 8^{-x} + 6$.

a What value is 8^{-x} always greater than?

b Hence, what value is $8^{-x} + 6$ always greater than?

c How many x-intercepts does $y = 8^{-x} + 6$ have?

d State the equation of the asymptote of the curve $y = 8^{-x} + 6$.

e State the domain.

f State the range.

14 Consider the function $y = 2\,(3^x)$.

a Find the y-value of the y-intercept.

b Can the function value ever be negative? Explain your answer.

c State the domain.

d State the range.

e As x approaches positive infinity, what value does y approach?

f Sketch the graph of the function.

15 Consider the function $y = -10^x$.

a Complete the following table:

x	-2	-1	0	1	2	3
y						

b For any value of x, is y always negative for this equation? Explain your answer.

c Sketch the graph of the function.

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16 Consider a graph of $y = 6^{3x}$.

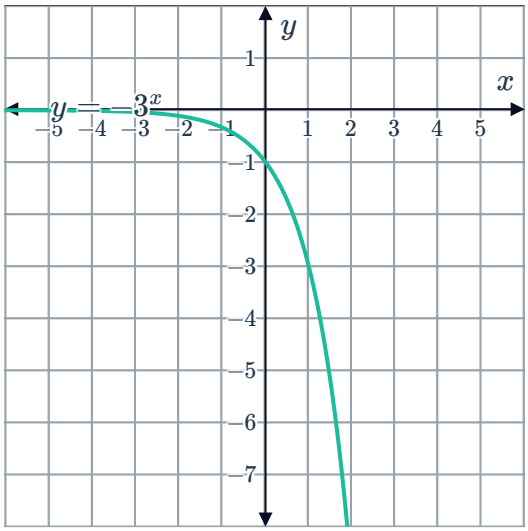
a A horizontal dilation by what factor takes the graph of $y = 6^x$ to $y = 6^{3x}$?

b Determine whether the following functions are equivalent to $y = 6^{3x}$:

i $y = 216 \times 6^x$ **ii** $y = \frac{1}{3} \times 6^x$ **iii** $y = 18^x$ **iv** $y = 216^x$

c Sketch the graph of $y = 6^{3x}$.

17 Consider the given graph of $y = -3^x$:



a State the asymptote of $y = -3^x$.

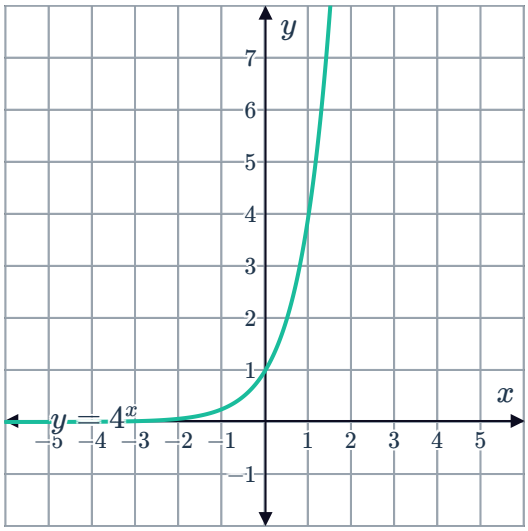
b Hence, find the asymptote of $y = 2 - 3^x$.

c How many x-intercepts would $y = 2 - 3^x$ have?

d State the domain of $y = 2 - 3^x$.

e State the range of $y = 2 - 3^x$.

18 Consider the given graph of $y = 4^x$:

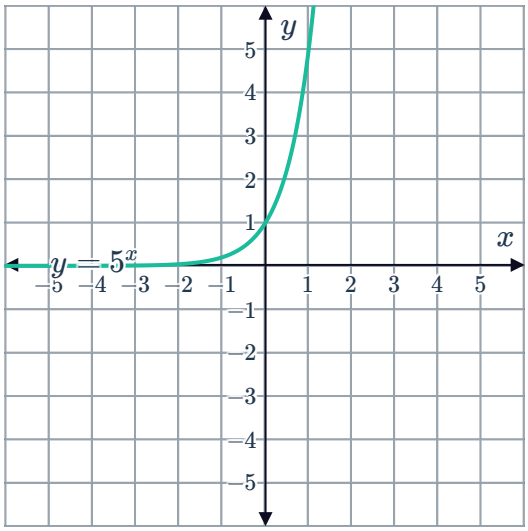


a Describe a transformation of the graph $y = 4^x$ that would obtain $y = 4^{-x}$.

b Sketch the graph of $y = 4^{-x}$.

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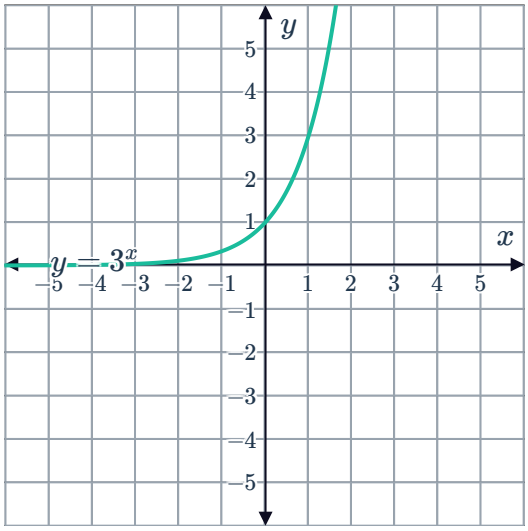
19 Consider the given graph of $y = 5^x$:



a Describe a transformation of the graph of $y = 5^x$ that would obtain $y = -5^x$:

b Sketch the graph of $y = -5^x$.

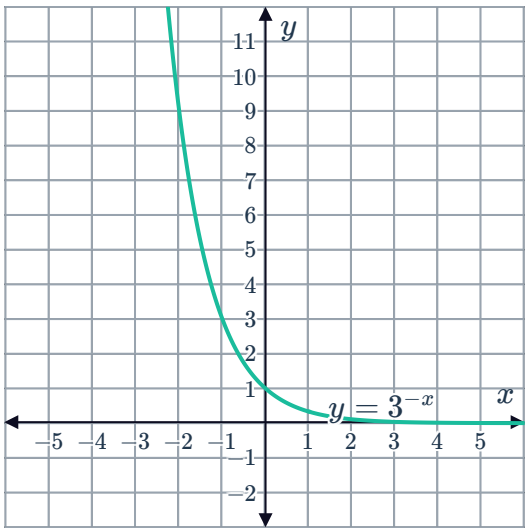
20 Consider the given graph of $y = 3^x$:



a Describe a transformation of the graph of $y = 3^x$ that would obtain $y = 3^x - 4$.

b Sketch the graph of $y = 3^x - 4$.

21 Consider the given graph of $y = 3^{-x}$:



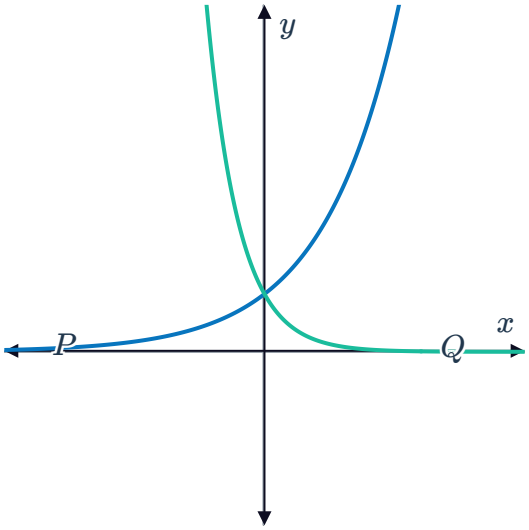
a Describe a transformation of the graph of $y = 3^{-x}$ that would obtain $y = 3^{-x} + 2$.

b Sketch the graph of $y = 3^{-x} + 2$.

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Function from a graph

22 Consider the given graphs of the two exponential functions P and Q:
State whether the following pairs of equations could be the equations of the graphs P and Q:



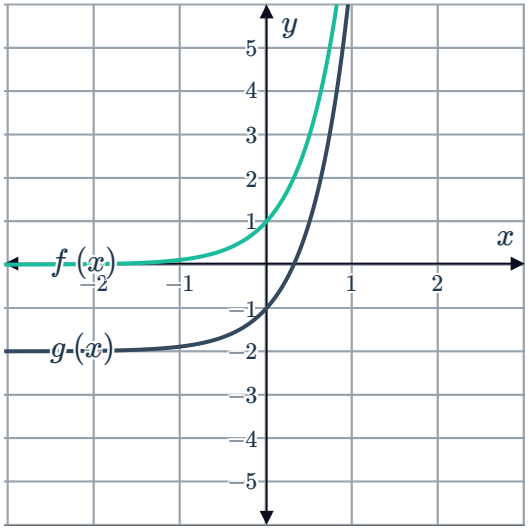
a $P: y = 2^x$
$Q: y = 2^{-x}$

b $P: y = (3.5)^x$
$Q: y = 6^{-x}$

c $P: y = 2^x$
$Q: y = 5^{-x}$

d $P: y = 5^x$
$Q: y = 2^{-x}$

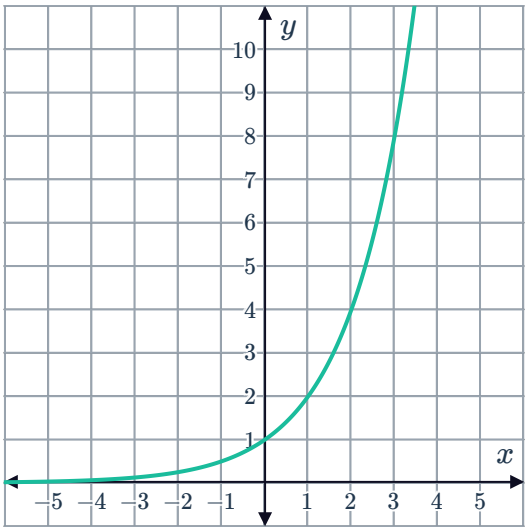
23 The graph of $f(x) = 9^x$ and another exponential function, $g(x)$ is shown:
$g(x)$ increasing at exactly the same rate as $f(x)$, but has a different y-intercept. Write down the equation of function $g(x)$.



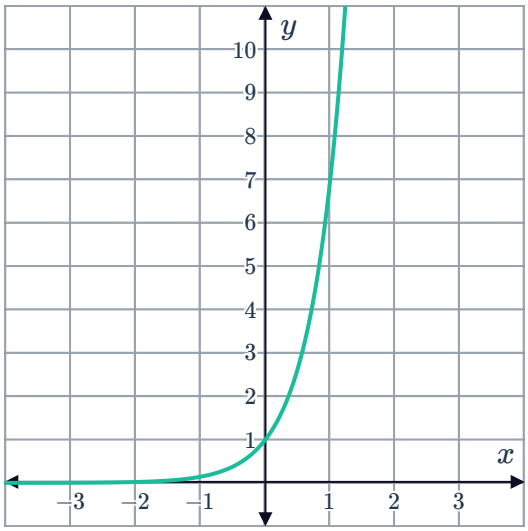
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24 For each of the following graphs of exponential functions in the form $y = a^x$, state the equation of the function:

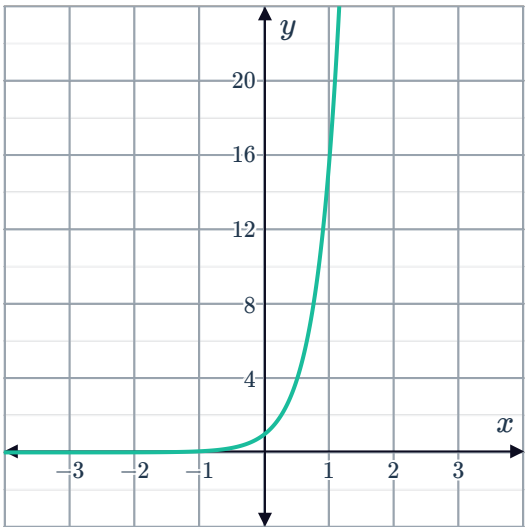
a



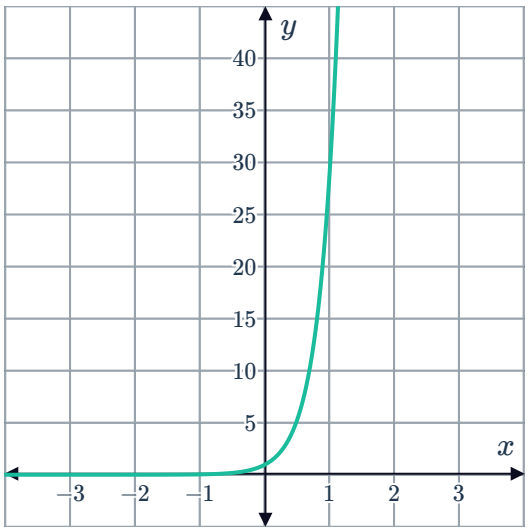
b



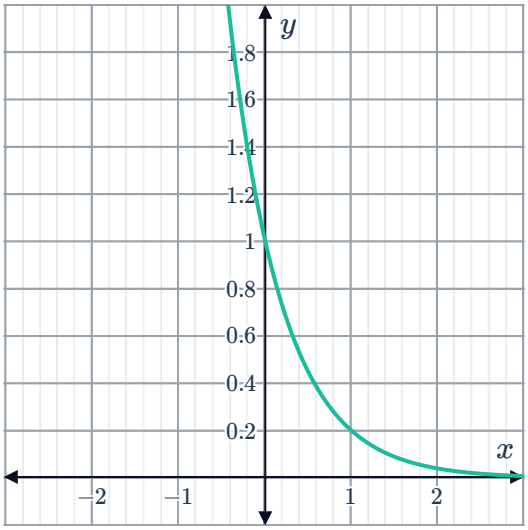
c



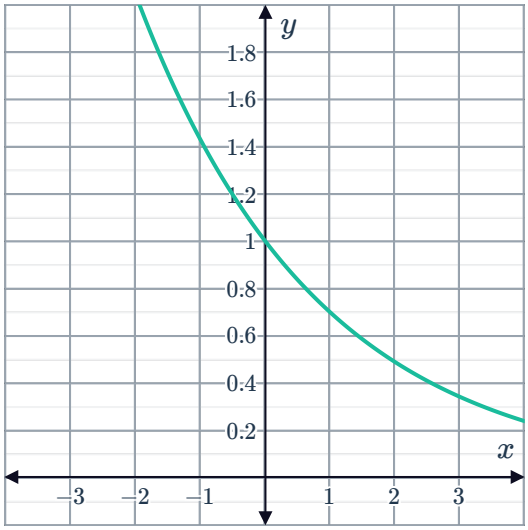
d



e



f



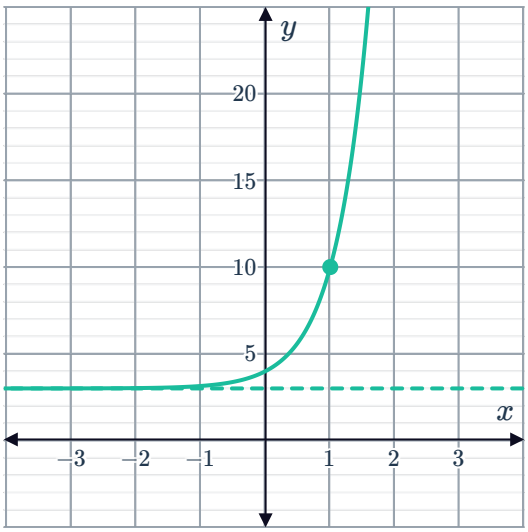
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25 For each of the following graphs of exponential functions in the form $y = a^x + k$:

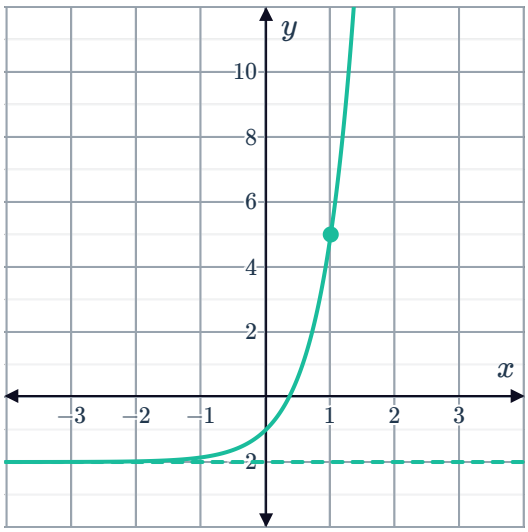
i State the equation for the horizontal asymptote.

ii State the equation for the exponential function.

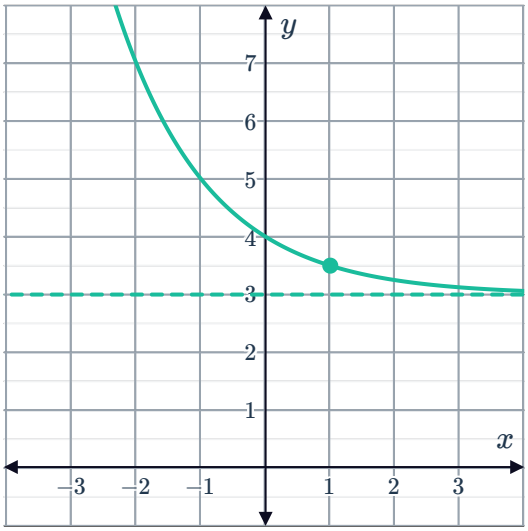
a



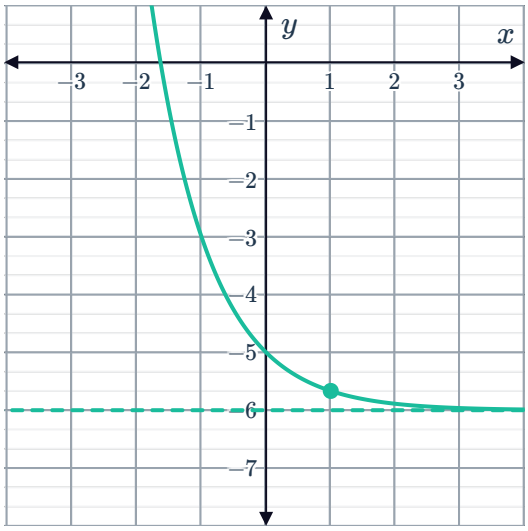
b



c



d



26 For each of the given pair of points, find the equation of the exponential function of the form $y = A \times a^x$ that passes through the points:

a $P\,(0, 3)$ and $Q\,(1, 6)$

b $P\,(1, 30)$ and $Q\,(2, 90)$

c $P\,(1, 20)$ and $Q\,(2, 5)$

d $P\,(1, 30)$ and $Q\,(2, 10)$

e $P\,(2, 20)$ and $Q\,(3, 40)$

f $P\,(2, -18)$ and $Q\,(5, -486)$

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